

Module 1 – Linux & Git

1. Linux Basics

What is Linux?

Linux is an open-source operating system widely used for servers, development, and DevOps environments.

Why Linux is Important

- Used in most servers
- Highly secure and stable
- Supports automation and scripting

Common Uses

- Web servers
 - Cloud systems
 - Development environments
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2. File System

What is File System?

The file system in Linux organizes and manages files and directories.

Key Directories

- /home – User files
- /etc – Configuration files
- /var – Logs and data
- /bin – System commands

Key Concepts

- Files and folders structure
 - Permissions
 - Path navigation
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3. Shell Commands

What is Shell?

Shell is a command-line interface used to interact with the system.

Common Commands

- `ls` → List files
- `cd` → Change directory
- `mkdir` → Create folder
- `rm` → Delete files
- `pwd` → Show current path

Example:

```
ls
cd folderName
mkdir newFolder
```

4. Git Basics

What is Git?

Git is a version control system used to track code changes.

Why Git?

- Tracks history
- Enables collaboration
- Manages versions

Basic Commands

- `git init` → Create repository
- `git add` → Stage changes
- `git commit` → Save changes

Example:

```
git init
git add .
git commit -m "Initial commit"
```

5. Branching

What is Branching?

Branching allows developers to work on different features independently.

Why Branching Matters

- Safe development
- Parallel work
- Easy collaboration

Common Commands

- git branch → Create branch
- git checkout → Switch branch
- git merge → Combine branches

Example:

```
git branch feature  
git checkout feature  
git merge feature
```

6. Key Concepts to Remember

- Linux is widely used in DevOps
 - File system organizes data
 - Shell commands control system
 - Git tracks code changes
 - Branching enables teamwork
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7. Summary

Linux and Git are fundamental tools in DevOps. Linux provides the environment for development and deployment, while Git ensures efficient version control and collaboration across teams.
